

Final Report

League of Legends Team Composition Diversity and Match Outcomes

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1. Introduction & Motivation

League of Legends is a competitive multiplayer online battle arena (MOBA) game in which two teams of five players compete to destroy the opposing team's base. Since each champion fulfills different strategic roles, team composition plays an important role in determining match dynamics and success.

I chose this project because League of Legends has been one of my favorite games for nearly ten years. Throughout this time, I became increasingly interested in how different team compositions affect match outcomes and whether balanced teams actually perform better statistically.

My friends and I often debated whether consistently balanced team compositions were always stronger, or whether compositions that became popular periodically and focused on specific roles were more advantageous. Therefore, in my first project on data analysis, I wanted to find data-driven answers to these questions I'd long wondered about.

In conclusion, this project examines the impact of role diversity in League of Legends team compositions on the probability of winning. The main goal is to investigate whether strategically more diverse and balanced team structures are more successful than teams with less diversity.

The project combines:

- exploratory data analysis (EDA),
- statistical hypothesis testing,
- and machine learning methods

to analyze the relationship between team composition and competitive success.

2. Dataset and Feature Engineering

The project uses ranked League of Legends match data obtained from Kaggle together with Riot Games champion metadata.

Data Sources

- `games.csv` → raw ranked match dataset including 51k ranked match from Europe server
- `champion_names&roles.json` → Riot API champion role metadata
- `team_role_dataset.csv` → processed team-level dataset created during the project

The raw dataset contains champion selections for both teams in each ranked match. However, champion IDs alone are not directly meaningful for strategic analysis.

To solve this problem, Riot's champion metadata was used to map each champion into a primary gameplay role:

- Marksman
- Tank
- Mage
- Assassin
- Support
- Fighter

After role mapping, feature engineering was performed to transform raw champion selections into structured numerical features.

For each team, the following variables were computed:

- number of Marksmen
- number of Tanks
- number of Mages
- number of Assassins
- number of Supports
- number of Fighters
- role diversity score

The diversity score represents the number of unique role categories present in a team composition.

Each match contributed two observations:

- one for the winning team
- one for the losing team

This produced a balanced team-level dataset suitable for statistical analysis and machine learning.

Additional engineered features were later added for machine learning experiments, including:

- high diversity indicator
- Marksman + Support synergy
- double Fighter compositions
- presence of Tanks

- presence of Supports

3. Exploratory Data Analysis (EDA)

The exploratory analysis focused on understanding how role diversity and composition structure affect win probability.

3.1 Win Rate vs Role Diversity

The first analysis examined how win rate changes as the number of unique roles in a team increases.

The results showed a clear positive relationship between diversity and win rate. Teams with 4–5 unique roles consistently achieved the highest win rates, while low-diversity teams underperformed.

These findings strongly suggested that balanced compositions provide strategic advantages.

3.2 Distribution of Diversity Levels

The distribution analysis showed that most ranked teams naturally fall into higher diversity categories.

This indicates that balanced compositions are already common in practical gameplay and are preferred by players.

3.3 Secondary Role Analyses

Additional analyses were conducted to test whether specific role structures independently improve performance.

The following structures were investigated:

- double Marksman teams
- Assassin-heavy compositions
- Tank-heavy compositions
- Marksman + Support synergy

Most of these analyses produced only moderate differences in win rate, generally remaining close to 50%.

This suggests that isolated role stacking alone is not a strong predictor of success.

However, Marksman + Support synergy produced slightly better outcomes, aligning with standard strategic gameplay patterns.

3.4 Winning Team Composition Analysis

The most common winning team compositions generally contained:

- one Marksman,
- one Mage,
- one Support,
- and multiple Fighters.

Low-diversity teams frequently over-stacked Fighters or Mages, while successful high-diversity teams maintained more balanced role distributions.

Overall, the EDA phase consistently suggested that:

overall composition diversity is more important than isolated role counts.

4. Hypothesis Testing

To statistically validate the EDA findings, a chi-square independence test was conducted between:

- role diversity score
- match outcome

Hypotheses

- **H₀:** Team role diversity and match outcomes are independent.
- **H₁:** Team role diversity and match outcomes are dependent.

Results

- Chi-square statistic = **54.18**
- p-value = **4.83e-11**
- Degrees of freedom = **4**

Since:

$p < 0.05$

the null hypothesis was rejected.

This confirms that team role diversity and match outcomes are statistically dependent.

Therefore, the statistical analysis validates the trends observed during exploratory data analysis.

5. Machine Learning Analysis

After completing the exploratory and statistical analyses, machine learning methods were applied to evaluate whether team composition features could predict match outcomes.

The objective was not only prediction accuracy, but also understanding which composition features contribute most strongly to winning probability.

5.1 Logistic Regression

Logistic Regression was used as an interpretable baseline classification model.

Results

- Accuracy $\approx$ **52%**
- Precision $\approx$ **52%**
- Recall $\approx$ **55%**
- F1-score $\approx$ **53%**

Although the predictive performance was moderate, the model successfully captured meaningful strategic patterns related to team composition.

The strongest positive coefficients were associated with:

- diversity,
- Marksman + Support synergy,
- balanced compositions.

These findings support the project's central hypothesis that overall composition structure matters more than isolated role stacking.

5.2 Random Forest

A Random Forest classifier was trained to capture non-linear relationships between role composition features and match outcomes.

Feature Importance Results

The most influential features were:

- Mage (0.149)
- Assassin (0.133)
- Fighter (0.123)

Interestingly, Logistic Regression coefficients for Mage and Assassin roles were negative, while Random Forest importance scores ranked them highly.

This suggests that their effects are non-linear:

- too many may damage balance,
- too few may reduce flexibility,
- balanced usage appears most effective.

Train and test accuracies remained close:

- Train Accuracy $\approx$ **51.9%**
- Test Accuracy $\approx$ **51.4%**

This indicates that severe overfitting was successfully avoided.

5.3 Interpretation of ML Results

Machine learning models achieved approximately 52% accuracy overall.

Although this may appear moderate, the result is expected because League of Legends outcomes depend on many external gameplay factors beyond team composition, including:

- player skill,
- macro gameplay,
- objective control,
- communication,
- execution quality.

Therefore, composition alone provides limited but still statistically meaningful predictive power.

6. Delta-Based Comparative Analysis

Instead of analyzing teams independently, an additional “delta-based” analysis was performed.

For each match:

Team A features – Team B features were computed.

This allowed the model to directly learn comparative composition advantages between opposing teams.

Results

The delta-based Logistic Regression model achieved performance similar to previous models.

The strongest positive coefficient was:

- delta diversity

This indicates that having higher diversity than the opposing team increases win probability.

Cross-validation scores also remained highly stable across folds:

- Mean CV Accuracy $\approx$ **51.7%**

This suggests that the model generalizes consistently despite moderate overall predictive accuracy.

7. Conclusions

This project investigated whether role diversity in League of Legends team compositions affects match outcomes.

Both statistical analysis and machine learning experiments consistently demonstrated that:

- higher diversity is associated with increased win probability,
- balanced compositions outperform heavily stacked teams,
- overall composition structure is more important than isolated role counts.

The chi-square hypothesis test statistically confirmed the relationship between diversity and match outcomes.

Machine learning analyses further supported these findings by identifying diversity and balanced structures as meaningful predictive features.

Overall, the findings strongly support the hypothesis that:

team composition diversity is a meaningful indicator of competitive success in League of Legends.

8. Limitations & Future Work

Although the project produced meaningful findings, several limitations remain.

Limitations

- Champion roles were simplified into primary categories only.
- Match outcomes are heavily influenced by player skill and in-game execution.
- External gameplay variables such as gold difference, objectives, and rank distribution were not included.
- The machine learning models used relatively simple composition-based features.

Future Work

Future improvements could include:

- rank-based analysis,
- temporal meta analysis,
- objective control statistics,
- advanced models such as XGBoost or neural networks,
- champion synergy network analysis.

9. AI Assistance Disclosure

AI tools (ChatGPT) were extensively used throughout the development of this project.

The model was primarily used for:

- generating and debugging Python code,
- assisting with machine learning implementations,
- structuring the Jupyter notebook,
- improving markdown formatting,
- interpreting statistical and machine learning outputs,
- and polishing the language of the final report.

Example prompts included:

- explanations for why machine learning accuracy remained moderate despite statistically meaningful findings,
- implementation support for Logistic Regression and Random Forest models,
- generation of markdown explanations for EDA and hypothesis testing sections,
- and interpretation of feature importance and coefficient analyses.

The generated outputs mainly consisted of:

- Python code snippets,
- debugging suggestions,
- markdown explanations,
- report text drafts,
- and interpretation support.

All project decisions, dataset selection, research direction, experimental design, and final evaluations were reviewed, modified where necessary, and approved by the author.